

NeuralKSP

AI-Powered Criminal Intelligence & Predictive Policing Platform

Executive Overview: Transforming static FIR databases into real-time bilingual actionable intelligence for field officers and command centers across Karnataka.

Key Capabilities: D3 Force-Directed Criminal Graph • Kingpin Centrality Scoring • Pathfinder Shortest Link Tracer • 12-Hour AM/PM Peak Crime Window Forecasting • 100% Kannada Script Transliteration

Resource / Attribute	Details & Deployment Link
■ Live Production URL	https://datathon-60080173971.development.catalystserverless.in/app/index.html
■ GitHub Repository	https://github.com/ramanathanmani/NeuralKSP
■■ Bilingual Support	English & Kannada (■■■■■) — 100% Transliteration across Nodes & Briefings
■■ Cloud Infrastructure	Zoho Catalyst Serverless Web Client (`client/` architecture)

1. Problem Statement & Existing Gaps in Policing

■ Fragmented FIR Data Silos Millions of FIRs and suspect profiles across 31 Karnataka districts stored in separate static databases make multi-district gang link analysis slow and manual.	■ Language & Accessibility Barrier Field constables and station officers across Karnataka require instant bilingual access in both English & Kannada script without typing complex database queries.
■ Manual Criminal Link Tracing Investigating multi-hop criminal syndicates relies on paper dossiers, taking days to identify ringleaders, gang associates, and hidden connection bridges.	■ Lack of Peak Window Forecasting Patrol force deployment relies on fixed shifts rather than predictive 12-hour AM/PM peak crime windows, reducing force effectiveness during high-risk hours.

2. Proposed Solution & NeuralKSP Modules

■ D3 Criminal Network Link Graph

Interactive 60 FPS physics graph mapping 300+ accused nodes, degree centrality kingpins (*Rekha Naik, Basavaraj Kumar*), and Pathfinder shortest link tracer (ACC-0004 → ACC-0018).

■ Predictive Crime Pulse Engine

D3 weekly rhythm heatmaps with 12-Hour AM/PM peak crime window forecasting (8:00 PM – 11:00 PM) and automated patrol force allocation directives.

■ Automated Station Briefings

Official confidential police intelligence briefings with 4 interactive tabs (Executive Summary, Threats, Patrol, Priority Watchlist) and 1-click PDF export.

■■ 100% Native Kannada Support (■■■■■)

Instant language switching for all metric cards, alert banners, tables, and full word-by-word Kannada transliteration for all 300 suspect names (■■■■■■■■■ ■■■).

3. System Architecture & Technology Stack

Component Layer	Technologies & Implementation Details
Frontend Stack	HTML5, ES6+ Vanilla JavaScript, Custom Glassmorphism CSS Token System
Data Visualization	D3.js v7 (Force-Directed Link Graphs & Weekly Heatmaps), Lucide Vector Icons
Serverless Cloud	Zoho Catalyst Web Client App Hosting (Deployed via `zcatalyst-cli`)
Quality Assurance	Playwright End-to-End Automated Browser Testing across Dark/Light & EN/KN modes
Performance	Zero heavy framework overhead, 60 FPS graph animation, offline-first client execution

4. Feasibility, Real-World KSP Impact & Scalability

■ 95% Faster Link Analysis

Reduces criminal syndicate discovery and gang leader profiling from days of paper dossier review to seconds of automated graph exploration.

■ Field Constable Ready

Zero learning curve; responsive slide drawer navigation works seamlessly on beat constable smartphones, tablets, and station workstations.

■ Zero Infra Overhead

Pure client-side rendering deployed on Zoho Catalyst serverless hosting eliminates expensive cloud database maintenance.

5. Working Prototype Verification & Submission Links

Resource	Live Link / Access Detail
■ Live Catalyst Production URL	https://datathon-60080173971.development.catalystserverless.in/app/index.html
■ GitHub Codebase Repository	https://github.com/ramanathanmani/NeuralKSP
■ Local Development Server	`py -3 -m http.server 8080` → `http://localhost:8080`
■ Verified Modules	Dashboard, Network Intel, Crime Pulse, AI Assistant, Briefings
■ Verified Languages	English & Kannada (■■■■■) — 100% Transliteration Verified